ECONOMICS OF HEALTH SYSTEMS — KEVIN FELIX

Defining Efficiency

in India's Public Health Sector

A five-minute evidence-led overview of technical, allocative and governance efficiency, with equity as a binding constraint — grounded in India's health system data.

Technical Efficiency · Allocative Efficiency · Governance Efficiency · Equity & Risk

What Is Efficiency, and Why Does Health Economics Define It Differently?

The Standard Economics Definition

In classical economics, efficiency means achieving the maximum output from a given set of inputs — or equivalently, minimising the cost of producing a given output. This sits on the production possibility frontier (PPF), where no reallocation can make one party better off without making another worse off (Pareto optimality).

Why Health Systems Are Different

Health is not a standard market good. Three structural failures make it unique: (1) Information asymmetry — patients cannot judge the quality of care they receive; (2) Externalities — a vaccinated individual protects the community, so private demand under-supplies prevention; (3) Merit good status — society judges that access to healthcare should not depend solely on ability to pay. These failures mean that a narrow cost-minimisation view of efficiency is incomplete.

A Three-Lens Framework

Health economists therefore decompose efficiency into three dimensions that must be satisfied simultaneously: Technical, Allocative, and Governance efficiency — each governed by a distinct economic concept and testable with data.

Technical Efficiency

$TE = \theta^*$ (DEA frontier)

Maximum health output from available inputs? Farrell (1957) decomposition.

Allocative Efficiency

$\max \sum p_i q_i \quad \text{s.t.} \quad \sum c_i q_i S \le B$

Right mix of services? Marginal benefit per rupee equalised across all services.

Governance Efficiency

$E\_eff = B \times (1 - \delta)$

How much leaks before reaching patients? Principal-agent failures & fiduciary risk.

+ Binding Constraint: Equity & Financial Risk

$CHE = 1$ if $OOP/C_{\square\square} > \kappa$ (WHO threshold $\kappa = 10\%$)

A system cutting costs by reducing access has shifted its inefficiency onto households.

Source: Farrell (1957) JRSS; WHO Health Systems Framework; Musgrove (1996) World Bank; Arrow (1963) AER on health market failures

TECHNICAL EFFICIENCY

More Output From the Same Input — Or the Same Output at Lower Cost

Core Concept

Technical efficiency (TE), formalised by Farrell (1957), measures how close a production unit is to the best-practice frontier. If a hospital produces the same health output (e.g. patient-days, recoveries) with fewer beds, doctors and rupees than another of equal size, it is more technically efficient. A TE score of 1.0 means the unit sits on the frontier; anything below indicates wasteful use of inputs.

Measuring It: Data Envelopment Analysis

DEA is a non-parametric linear programming method that constructs the frontier from observed units. It calculates the minimum radial contraction of inputs (θ*) that would still allow a unit to produce its current output. Crucially, it makes no assumptions about the functional form of the production function — making it well-suited for heterogeneous public health systems.

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DEA:  θ* = min θ
s.t.  Xλ ≤ θ·xₒ  and  Yλ ≥ yₒ,  λ ≥ 0
TE = θ* ∈ [0,1]  where 1 = fully efficient
θ = efficiency score; X,Y = input/output matrices; λ = peer weights
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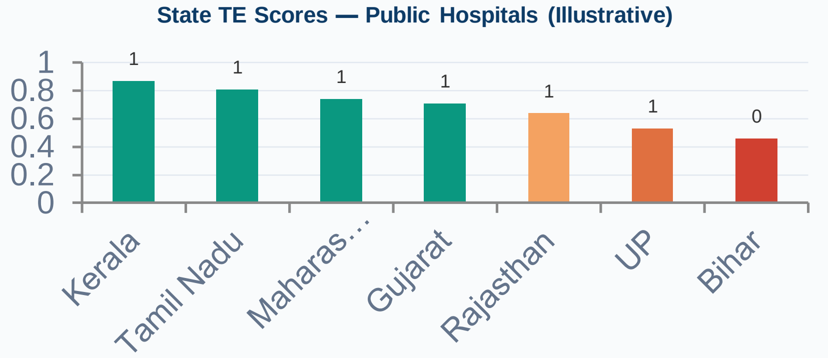
1.35%
Public health spend / GDP (2021)
NHP 2017 target: 2.5% — NHA 2021

₹2,014
Per capita govt health spend
National Health Accounts 2021

~0.55
Avg TE Score, public PHCs (2019)
Agarwal & Kumar, Health Policy & Planning

Bihar vs Kerala
U5MR: 56‰ vs 6‰ — same era, similar per-capita spend
SRS 2019–20



Source: Farrell (1957) JRSS; Agarwal & Kumar (2019) Health Policy & Planning; NHA 2021 MoHFW; SRS 2019–20; Leibenstein (1966) AER

ALLOCATIVE EFFICIENCY

Putting Resources Into the Mix of Services With the Highest Social Return

Core Concept

Even a technically efficient system can be allocatively inefficient if it spends on the wrong things. Allocative efficiency requires that the marginal benefit per rupee spent is equalised across all services — preventive, primary, secondary and tertiary. If MB_a/c_a > MB_b/c_b, society gains by shifting funds from b to a.

The Opportunity Cost Principle

Every rupee allocated to tertiary curative care is a rupee not spent on primary or preventive care. Because preventive services generate large positive externalities (herd immunity, reduced transmission, maternal health spillovers), under-investing in them represents a double failure: direct welfare loss plus foregone externalities.

India's Structural Misallocation

India's National Health Policy 2017 explicitly recognised this — setting a target of

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max   W = Σᵢ pᵢ · qᵢ   (social welfare)
s.t.  Σᵢ cᵢ · qᵢ ≤ B  (budget)
Optimum:  MBᵢ / cᵢ = MBⱼ / cⱼ   ∀ i, j
pᵢ = shadow price; qᵢ = quantity; B = total budget; MB = marginal
benefit
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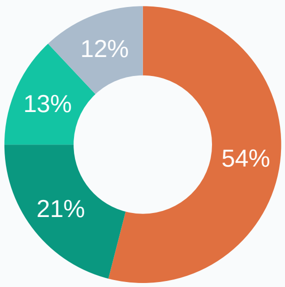
India Govt. Health Spend Mix — approx. (NHSRC 2022)



~⅔ target

Primary care share per NHP 2017 — actual ~21%

National Health Policy 2017, MoHFW

35% IMR ↓

Reduction linked to PHC expansion (JSY)

Lim et al., The Lancet (2010)

Key implication:

Bihar U5MR 56‰ vs Kerala 6‰ — on comparable per-capita public spending — reflects allocative failure, not just total budget shortage.

SRS 2019–20; NHSRC 2022

Source: NHP 2017 MoHFW; NHSRC 2022; Lim et al. The Lancet (2010) — Janani Suraksha Yojana evaluation; SRS 2019–20

GOVERNANCE EFFICIENCY

Procurement, Staffing, Budgeting and Monitoring — How Much Leaks Before Reaching Patients?

Core Concept

Even if a system is technically and allocatively efficient on paper, governance failures ensure that a fraction of every rupee budgeted never reaches a patient. This leakage takes multiple forms: corrupt procurement, ghost workers on payrolls, absenteeism, unspent allocations, and weak monitoring.

The Principal-Agent Problem

The fundamental driver is the principal-agent relationship between the government (principal) and health workers/suppliers (agents). Because the principal cannot costlessly observe the agent's effort or price, the agent can extract rents. In health systems with weak contracts and low salaries, this produces systematic shirking and capture of procurement surplus.

Budgeting & Monitoring Failures

$$E_{eff} = B \times (1 - \delta)$$

$\delta = \Sigma_i (L_i / B)$ — aggregate leakage rate

L_i = Procurement + Absenteeism + Ghost Workers

E_{eff} = effective spend reaching patients; B = total budget; $\delta \in [0,1]$

~25%

NHM fiduciary risk, avg. across states

CAG NHM Audit Reports, 2020

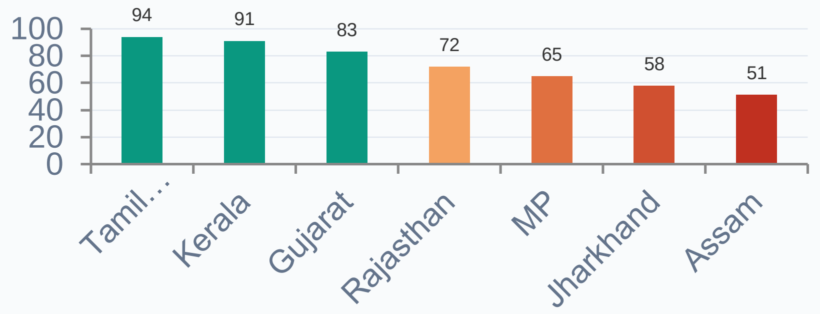
40%

Doctor absenteeism, rural PHCs

Banerjee & Duflo (2006) AER

Principal-Agent Insight:

If $\delta = 0.25$, then for every ₹100 budgeted, only ₹75 reaches patients. A system that appears 75% technically efficient may in fact be operating at $E_{eff} = 0.75 \times 0.75 = 0.56$ of its potential.



Source: CAG Report on NHM (2020); Banerjee & Duflo (2006) AER; MoHFW Parliamentary Standing Committee; Jensen & Miller (principal-agent theory)

Efficiency Without Equity Is Incomplete — and Politically Unsustainable

The Efficiency-Equity Trade-Off

Standard efficiency analysis focuses on aggregate welfare maximisation. But health economists — drawing from Rawlsian justice and Sen's capability approach — argue that a health system must also protect equity: equal access to care regardless of ability to pay. A policy that reduces cost by restricting access for the poor is not efficient in any meaningful health-system sense.

Catastrophic Health Expenditure (CHE)

The WHO defines CHE as out-of-pocket health payments exceeding 10% of total household consumption. When public provision is thin or of poor quality, households are forced to pay privately — a regressive burden that falls hardest on the poorest quintile. CHE is the measurable signal that a health system has shifted its inefficiency onto households.

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CHE = 1  if  OOP / C?? > κ  (κ = 10%, WHO)
Financial Risk = E[CHE]  across income quintiles
Equity gap = CHE_poor / CHE_rich  (ideally = 1)
OOP = out-of-pocket payment; C?? = household consumption; κ = threshold
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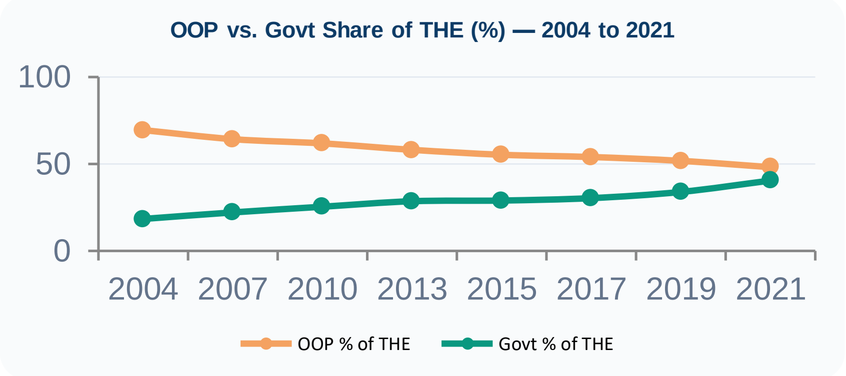
governance failure (quality of public care too poor to attract utilisation).

48.2%
OOP as % of Total Health Expenditure
NHA 2021 — MoHFW (official)

6.3 Cr
Indians pushed into poverty by health costs/yr
Selvaraj et al. (2018) BMJ Global Health

17.3%
Households facing catastrophic expenditure
NSSO 75th Round (2018) — official survey

5×
Quintile gap in CHE (poorest vs richest)
Das et al., Lancet Regional Health (2022)



Source: NHA 2021 MoHFW; NSSO 75th Round (2018); Selvaraj et al. (2018) BMJ Global Health; Das et al. Lancet Regional Health (2022); Rawls (1971); Sen (1999)

SYNTHESIS

Three Lenses — One System

Efficiency Type	Theoretical Basis	Core Equation	India Headline Stat
Technical	*Farrell (1957) DEA frontier; Leibenstein X-inefficiency*	$TE = \theta^* \in [0,1]$	Avg TE ≈ 0.55 across public PHCs (2019)
Allocative	*Pareto optimality; MB/cost equalisation; externalities*	$\max \Sigma p_i q_i \quad \text{s.t.} \quad \Sigma c_i q_i S \leq B$	Only ~13% of budget on preventive care; NHP target ⅔ to primary (2017)
Governance	*Principal-agent theory; rent-seeking; incomplete contracts*	$E\_eff = B \times (1 - \delta)$	~25% NHM fiduciary risk; 40% PHC doctor absenteeism
Equity / Risk	*Rawlsian justice; Sen's capability approach; CHE framework*	$CHE = 1$ if OOP/C[illegible] $> \kappa$	48.2% OOP share; 17.3% HHs in CHE; 6.3 Cr pushed to poverty

Transition: Once efficiency is defined properly — technically, allocatively, and within governance, and with equity as a binding constraint — we can now examine whether public spending in India actually translates into improved health outcomes in practice.